INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

THE SITE

Location	Centersquare Lynnwood SE1, WA -- station KPAE, America/Los_Angeles
Building	OSM 396296733 Centersquare Lynnwood SE1
Facade gap	not applicable -- one building, no second facade
Weather record	43,618 real hourly records from KPAE
Plume physics	NOT MODELLED. No other tagged data centre lies inside the solver's validated range, so there is no neighbour intake for a plume to arrive at: the quantity does not exist here rather than being unmeasured. This is a statement about the model's domain, not a claim that recirculation here is zero. A building's own exhaust re-entering its own intake is not modelled at any site in this project.

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2025-09-16 (crossing)
Plant limit	21.0 C
Notice required	3 h
Forecast skill	0.50 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	2 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 10 of 24 hours with 2 mode changes. The reactive incumbent took 12 hours -- 2 more -- but declared 3 unsafe hours against the agent's 1.

Agent	10 of 24 hours free cooling, 2 mode change(s), 1 unsafe hour(s)
Incumbent	12 of 24 hours free cooling, 2 mode change(s), 3 unsafe hour(s)

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	FREE	yes	-	13.055	21.0	12.780	0.709
01	FREE	yes	-	13.330	21.0	13.330	0.646
02	FREE	yes	-	12.775	21.0	12.220	0.614
03	FREE	yes	-	12.775	21.0	12.220	0.589
04	FREE	yes	-	13.055	21.0	12.220	0.555
05	FREE	yes	-	12.500	21.0	12.220	0.500
06	FREE	yes	-	12.775	21.0	12.220	0.614
07	FREE	yes	-	14.165	21.0	13.890	0.905
08	FREE	yes	-	15.830	21.0	16.110	1.078
09	FREE	yes	-	18.885	21.0	21.110	1.288
10	mechanical	no	dry-bulb	21.115	21.0	23.890	1.062

11	mechanical	no	dry-bulb	22.500	21.0	24.440	1.011
12	mechanical	no	dry-bulb	25.830	21.0	26.110	1.081
13	mechanical	no	dry-bulb	28.055	21.0	28.330	0.950
14	mechanical	no	dry-bulb	28.335	21.0	28.890	0.907
15	mechanical	no	dry-bulb	30.000	21.0	31.110	0.928
16	mechanical	no	dry-bulb	30.830	21.0	31.110	1.055
17	mechanical	no	dry-bulb	30.280	21.0	30.560	0.893
18	mechanical	no	dry-bulb	30.275	21.0	29.440	0.715
19	mechanical	no	dry-bulb	29.165	21.0	27.220	0.996
20	mechanical	no	dry-bulb	28.335	21.0	26.110	1.122
21	mechanical	no	dry-bulb	27.500	21.0	25.560	1.014
22	mechanical	no	dry-bulb	25.830	21.0	24.440	0.772
23	mechanical	no	dry-bulb	25.275	21.0	23.890	0.815

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 13.055 C, which is 7.945 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.709 C, and the margin is measured, not chosen: 0.709 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

01:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 13.330 C, which is 7.670 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.646 C, and the margin is measured, not chosen: 0.646 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

02:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 12.775 C, which is 8.225 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.614 C, and the margin is measured, not chosen: 0.614 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

03:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 12.775 C, which is 8.225 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.589 C, and the margin is measured, not chosen: 0.589 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

04:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 13.055 C, which is 7.945 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.555 C, and the margin is measured, not chosen: 0.555 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

05:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 12.500 C, which is 8.500 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.500 C, and the margin is measured, not chosen: 0.500 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

06:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 12.775 C, which is 8.225 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.614 C, and the margin is measured, not chosen: 0.614 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

07:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 14.165 C, which is 6.835 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.905 C, and the margin is measured, not chosen: 0.905 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

08:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.830 C, which is 5.170 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.078 C, and the margin is measured, not chosen: 1.078 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

09:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.885 C, which is 2.115 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.288 C, and the margin is measured, not chosen: 1.288 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by. *** THIS WAS WRONG: the intake actually reached 21.110 C, above the limit. The bound failed here, and it is counted as a breach rather than explained away. ***

10:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.115 C against a 21.0 C limit, so it fails by 0.115 C. It would take a limit 0.115 C higher, or a bound 0.115 C tighter, to change this hour.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.500 C against a 21.0 C limit, so it fails by 1.500 C. It would take a limit 1.500 C higher, or a bound 1.500 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.830 C against a 21.0 C limit, so it fails by 4.830 C. It would take a limit 4.830 C higher, or a bound 4.830 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 28.055 C against a 21.0 C limit, so it fails by 7.055 C. It would take a limit 7.055 C higher, or a bound 7.055 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 28.335 C against a 21.0 C limit, so it fails by 7.335 C. It would take a limit 7.335 C higher, or a bound 7.335 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 30.000 C against a 21.0 C limit, so it fails by 9.000 C. It would take a limit 9.000 C higher, or a bound 9.000 C tighter, to change this hour.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 30.830 C against a 21.0 C limit, so it fails by 9.830 C. It would take a limit 9.830 C higher, or a bound 9.830 C tighter, to change this hour.

17:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 30.280 C against a 21.0 C limit, so it fails by 9.280 C. It would take a limit 9.280 C higher, or a bound 9.280 C tighter, to change this hour.

18:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 30.275 C against a 21.0 C limit, so it fails by 9.275 C. It would take a limit 9.275 C higher, or a bound 9.275 C tighter, to change this hour.

19:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 29.165 C against a 21.0 C limit, so it fails by 8.165 C. It would take a limit 8.165 C higher, or a bound 8.165 C tighter, to change this hour.

20:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 28.335 C against a 21.0 C limit, so it fails by 7.335 C. It would take a limit 7.335 C higher, or a bound 7.335 C tighter, to change this hour.

21:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 27.500 C against a 21.0 C limit, so it fails by 6.500 C. It would take a limit 6.500 C higher, or a bound 6.500 C tighter, to change this hour.

22:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.830 C against a 21.0 C limit, so it fails by 4.830 C. It would take a limit 4.830 C higher, or a bound 4.830 C tighter, to change this hour.

23:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.275 C against a 21.0 C limit, so it fails by 4.275 C. It would take a limit 4.275 C higher, or a bound 4.275 C tighter, to change this hour.

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